

Michael J. Arcaro

Department of Psychology
University of Pennsylvania

marcaro@sas.upenn.edu
<https://arcarolab.org>

EMPLOYMENT

- 2020 – Assistant Professor
Department of Psychology
University of Pennsylvania
- 2017 – 2019 Instructor in Neurobiology
Advisor: Margaret Livingstone, Ph.D.
Livingstone Laboratory, Harvard Medical School
- 2015 – 2017 Postdoctoral Fellow
Advisor: Margaret Livingstone, Ph.D.
Livingstone Laboratory, Harvard Medical School
- 2013 – 2015 Postdoctoral Research Associate
Advisor: Sabine Kastner, Ph.D., M.D.
Neuroscience of Attention & Perception Laboratory, Princeton University

EDUCATION

- 2013 Princeton University – Ph.D. in Psychology and Neuroscience
- 2011 Princeton University – M.A. in Psychology and Neuroscience
- 2004 Boston University – B.A. in Psychology and Philosophy (*cum laude*)

AWARDS & HONORS

- 2023 APA's Distinguished Scientific Award for Early Career Contribution to Psychology
- 2019 David Hubel Outstanding Postdoctoral Fellow
- 2018 Harvard Faculty Research Award – Mind, Brain, and Behavior program
- 2017 William Randolph Hearst Fellowship
- 2016 Mahoney Postdoctoral Fellow
- 2010 Quantitative and Computational Neuroscience Fellowship / NIH T90

PREPRINT

Meyer EE, Ong WS, Song C, Cottaris NP, Zhang L, Collina J, Brainard DH, **Arcaro MJ**. Transformation-tolerant object recognition in tree shrews despite lacking a fovea. *bioRxiv*.

Kim H, **Arcaro MJ***, Imam N*. Geometric constraints in the development of primate extrastriate visual cortex. *bioRxiv*.

Oishi H, Berezovskii VK, Livingstone MS, Weiner KS, **Arcaro MJ**. Metabolic organization of macaques visual cortex reflects retinotopic eccentricity and category selectivity. *bioRxiv*.

Ellwood-Lowe ME, Nishio M, Dufford AE, Arcaro MJ, Satterthwaite TD, Mackey. Cascading periods of language-related brain plasticity across early childhood. *bioRxiv*.

PEER-REVIEWED PUBLICATIONS

Ayzenberg V & **Arcaro MJ** (*in press*) The neural correlates of core perceptual capacities in infants. *Brain and Behavioral Sciences*.

Ayzenberg V & **Arcaro MJ** (2026) Functional organization of the human visual system at birth and across late gestation. *Neuron*.

Nishio M, Wang X, Cornblath EJ, Lee SH, Shih YI, Palomero-Gallagher N, **Arcaro MJ**, Lydon-Staley DM, Mackey AP. (2026) Alcohol impacts an fMRI marker of neural inhibition in humans and rodents. *NeuroImage*.

Guest DR, Allen EJ, Kay MN*, **Arcaro MJ*** (2026) Processing of natural scenes in the human pulvinar. *Nature Communications*.

Nishio M, Liu X, Mackey AP, **Arcaro MJ** (2025) Humans have a longer period of cortical maturation across depth and hierarchy than macaques. *PLOS Biology*.

Sydnor VJ, Bagautdinova J, Larsen B, **Arcaro MJ**, Barch DM, Bassett DS, Alexander-Bloch AF, Cook PA, Covitz S, Franco AR, Gur RE, Gur RC, Mackey AP, Mehta K, Meisler SL, Milham MP, Moore TM, Müller EJ, Roalf DR, Salo T, Schubiner G, Seidlitz J, Shinohara RT, Shine JM, Yeh FC, Cieslak M, Satterthwaite TD. (2025) A sensorimotor-association axis of thalamocortical connection development. *Nature Neuroscience*.

Ayzenberg V, Song L, **Arcaro MJ** (2025) An intrinsic hierarchical, retinotopic organization of pulvinar connectivity in the human neonate. *Current Biology*.

Meyer E, Martynek M, Kastner S, Livingstone MS, **Arcaro MJ** (2025) Expansion of a conserved architecture drives the evolution of primate visual cortex. *PNAS*.

Oishi H, Berezovskii VK, Livingstone MS, Weiner KS*, **Arcaro MJ*** (2024) Inferotemporal face patches are histo-architecturally distinct. *Cell Reports*.

Bourne JA, Cichy RM, Kiorpes L, Morrone MC, **Arcaro MJ**, Nielsen KJ. (2024) Development of higher-level vision: a network perspective. *Journal of Neuroscience*.

Arcaro MJ & Livingstone MS (2024). A whole-brain topographic ontology. *Annual Reviews Neuroscience*.

Linsely D, Rodriguez IF, Fei T, **Arcaro MJ**, Sharma S, Livingstone MS, Serre T (2023). Performance-optimized deep neural networks are evolving into worse models of inferotemporal cortex. *Advances in Neural Information Processing Systems*.

Kay K, Bonnen K, Denison RN, **Arcaro MJ**, Barack DL (2023). Tasks and their role in visual neuroscience. *Neuron*.

Scott LS & **Arcaro MJ** (2023). A domain-relevant framework for the development of face processing. *Nature Reviews Psychology*. 2, 183-195.

Ellis CT, Yates TS, **Arcaro MJ**, Turk-Browne NB (2023). Movies reveal the fine-grained organization of infant visual cortex. *eLife*.

Arcaro MJ, Livingstone MS, Kay KN, Weiner KS (2022). The retrocalcarine sulcus is functionally distinct between macaques and humans. *Brain Structure and Function*, 227(4), 1227-1245.

Arcaro MJ and Livingstone MS (2021) On the relationship between maps and domains in inferotemporal cortex. *Nature Reviews Neuroscience*. 22, 573-583.

Ellis CT, Yates TS, Skalaban LJ, Bejjanki VR, **Arcaro MJ**, Turk-Browne NB (2021). Retinotopic organization of visual cortex in human infants. *Neuron*. 109 (16), 2616-2626.

Natu V, **Arcaro MJ**, Barnett MA, Gomez J, Livingstone MS, Grill-Spector K, Weiner KS (2021) Sulcal depth in medial ventral temporal cortex predicts the location of a place-selective region in macaques, children, and adults. *Cerebral Cortex*, 31. 48-61.

Arcaro MJ, Mautz T, Berezhovskii V, Livingstone MS (2020) Anatomical correlates of face patches in macaque inferotemporal cortex. *PNAS*. 117 (51), 32667-32678.

Arcaro MJ, Ponce CR, Livingstone MS (2020) The neurons that mistook a hat for a face. *eLife*. 9(e53798), 1-19.

Arcaro MJ, Schade P, Livingstone MS (2019) Body-map proto-organization in newborn macaques. *PNAS*. 116(49) 24861-24871.

Arcaro MJ, Schade PF, Livingstone MS (2019) Universal mechanisms and the development of the face network: what you see is what you get. *Annual Review of Vision Science*. 5. 341-372.

Livingstone MS, **Arcaro MJ**, Schade P (2019) Cortex is cortex: ubiquitous principles drive face-domain development. *Trends in Cognitive Sciences*. 23 (1), 3-4.

Arcaro MJ, Thaler L, Quinlan DJ, Monaco S, Khan S, Valyear KF, Goebel R, Dutton GN, Goodale MA, Kastner S, Culham JC. (2019) Psychophysical and neuroimaging responses to moving stimuli in a patient with the Riddoch phenomenon due to bilateral visual cortex lesions. *Neuropsychologia*. 128. 150-165.

Benson N, Jamison KW, **Arcaro MJ**, Vu A, Glasser MF, Coalson TS, Van Essen D, Yacoub E, Ugurbil K, Winawer J, Kay K. (2018) The HCP 7T Retinotopy Dataset: Description and pRF Analysis. *Journal of Vision*. 18 (13) 1-22.

Arcaro MJ, Pinsk MA, Chen J, Kastner S. (2018) Organizing principles of pulvino-cortical coupling in humans. *Nature Communications*. 9(1), 1-14.

Haufe S, DeGuzman P, Henin S, **Arcaro MJ**, Honey CJ, Hasson U, Parra LC. (2018) Elucidating relations between fMRI, ECoG and EEG through a common natural stimulus. *NeuroImage*. 179, 79-91.

Todd N, Zhang Y, **Arcaro MJ**, Becerra L, Borsook D, Livingstone MS, McDannold N. (2018) Focused ultrasound induced opening of the blood-brain barrier disrupts inter-hemispheric resting state functional connectivity in the rat brain. *NeuroImage*. 178, 414-422.

Arcaro MJ*, Schade PF*, Vincent JL, Ponce CR, Livingstone MS*. (2017) Seeing faces is necessary for face-patch formation. *Nature Neuroscience*. 20(10), 1-9.

Arcaro MJ & Livingstone MS. (2017) A hierarchical, retinotopic proto-organization of the primate visual system at birth. *eLife*. 6(e26196), 1-24.

Arcaro MJ & Livingstone MS. (2017) Retinotopic organization of scene areas in the macaque inferior temporal cortex. *Journal of Neuroscience*. 37(31), 7373-7389.

Livingstone MS*, Vincent JL*, **Arcaro MJ***, Srihasam K, Schade P, Savage T. (2017) Development of the macaque face-patch system. *Nature Communications*. 8, 1-12.

Chen J, Honey CJ, Simony E, **Arcaro MJ**, Norman KA, Hasson U. (2016) Accessing real-life episodic information from minutes versus hours earlier modulates hippocampal and high-order cortical dynamics. *Cerebral Cortex*. 28(8), 3428-3441.

Arcaro MJ, Pinsk MA, Kastner S. (2015) The anatomical and functional organization of the human visual pulvinar. *Journal of Neuroscience*. 35(27), 9848-9871.

Arcaro MJ & Kastner S. (2015) Topographic organization of areas V3 and V4 and its relation to supra-areal organization of the primate visual system. *Visual Neuroscience*. 32(e015), 1-15.

Arcaro MJ, Honey CJ, Mruczek REB, Kastner S, Hasson U. (2015) Widespread correlation patterns of fMRI signal across visual cortex reflect eccentricity organization. *eLife*. 4(e03952), 1-28.

Lombaert H, **Arcaro MJ**, Ayache N. (2015) Brain transfer: spectral analysis of cortical surfaces and functional maps. *IPMI*. 9123, 474-487.

Wang L, Mruczek REB, **Arcaro MJ**, Kastner S. (2015) Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*. 25(10) 3911-3931.

Kelly YT, Webb TW, Meier JD, **Arcaro MJ**, Graziano MSA (2014). Attributing awareness to oneself and to others. *PNAS*. 111(13), 5012-5017.

Wang L, Saalmann YB, Pinsk MA, **Arcaro MJ**, Kastner S (2012). Electrophysiological low-frequency coherence and cross-frequency coupling contributes to BOLD connectivity. *Neuron*. 76(5), 1010-1020.

Arcaro MJ, Pinsk MA, Li X, Kastner S (2011). Visuotopic organization of macaque posterior parietal cortex: An fMRI study. *Journal of Neuroscience*. 31(6), 2064-2078.

Caplovitz GP, **Arcaro M**, Kastner S (2010). Stage 3 and what we see. *Cognitive Neuroscience*. 1(3), 220-222.

Carmel D, **Arcaro MJ**, Kastner S, Hasson U (2010). How to create and use binocular rivalry. *Journal of Visualized Experiments (JoVE)*. 45(e2030), 1-10.

Arcaro MJ*, McMains S*, Singer B, Kastner S (2009). Retinotopic organization of human ventral visual cortex. *Journal of Neuroscience*. 29(34), 10638-10652.

Pinsk MA, **Arcaro M**, Weiner KS, Kalkus JF, Inati SJ, Gross CG, Kastner S (2009). Neural representations of faces and body parts in the macaque and human cortex: A comparative fMRI study. *Journal of Neurophysiology*. 101, 2581-2600.

BOOK CHAPTERS

Weiner KS, Willbrand EH, Maboudian SA, Oishi H, Natu VS, Amiez C, **Arcaro MJ**. (*in press*) The cortical morphology and microarchitecture of face processing in primates. *Oxford Handbook of Face Perception*.

Arcaro MJ, Willbrand EH, Amiez C, Maboudian SA, Weiner KS. (2025). Evolution of functional sulcal anatomy in primates. In Kass J.H. &erculano-Houzel S., editor. *The Evolution of the Nervous Systems of Nonhuman Primates*. 3rd Edition.

Kastner S & **Arcaro MJ**. (2022) The Thalamus in Attention. In Halassa M.M., editor. *The Thalamus*. Cambridge, UK. Cambridge University Press.

PROFESSIONAL ACTIVITIES AND SERVICE

- 2024 Co-chair for Society for Neuroscience minisymposium, *Development of Higher-Level Vision*
- 2021 Session moderator for Vision Sciences Society talk session: *Development*
- 2018 Co-chair for Society for Neuroscience nanosymposium, *Vision: Representation of Faces and Bodies*
- 2017 Postdoctoral steering committee for Harvard's Mind Brain Behavior program

TEACHING

- 2022- PSYC 1230 Cognitive Neuroscience - Instructor
- 2020- PSYC 3233 Seminar in Cognitive Neuroscience: Brain Development - Instructor
- 2014 NEU 502 From Molecules to Systems to Behavior - Workshop: Functional connectivity
- 2013 PSY 255 Cognitive Psychology – Precept Instructor
- 2012 PSY 311 Rationality and Human Reasoning – Precept Instructor
- 2011 PSY 259 Cognitive Neuroscience - Lab Instructor
- 2010 NEU 502 From Molecules to Systems to Behavior - Workshop: Retinotopic mapping with fMRI

MENTORING

Postdoctoral Fellows

- 2023- Xingyu Liu, Postdoctoral Researcher
- 2022-2025 Vladislav Ayzenberg, Postdoctoral Researcher
Assistant Professor, Temple University
Data-Driven Discovery Initiative Postdoctoral Fellow
MindCORE Postdoctoral Research Fellowship
- 2022-2025 Hiroki Oishi, Postdoctoral Researcher (co-advised w/ Kevin Weiner, UC Berkeley)
Assistant Professor, National Institute of Physiological Sciences, Japan
- 2020-2021 Wei Song Ong, Postdoctoral Researcher, University of Pennsylvania

Graduate Students

- 2021- Emily Meyer, Graduate Student
2023 Vision Sciences Society Travel Award
- 2024- Monami Nishio, Graduate Student (primary advisor: Allyson Mackey)
- 2025- Diana King, Graduate Student
- 2025- Reyansh Sathishkumar, Graduate Student
- 2021-2023 Lucy Song, Graduate Student

Undergraduates

- 2026- Jacob Tuteur, University of Pennsylvania
2026 MindCORE Undergraduate Summer Research Fellowship
- 2024- Anna Lugovskoy, University of Pennsylvania
2024 Ruth Marcus Kanter College Alumni Society Undergraduate Research Grant
2025 Tree Shrew Discovery Conference Travel Grant (1st author poster presentation)
- 2024-2026 Jessica Wu, University of Pennsylvania
2024 MindCORE Undergraduate Summer Research Fellowship
2025 Penn Neuroscience Program Fellowship for Summer Research
2026 Eliot Stellar Thesis Prize
- 2025-2026 Nate Sirlin, University of Pennsylvania
- 2025-2026 Utkarsh Tannan, University of Pennsylvania
2025 MindCORE Undergraduate Summer Research Fellowship
- 2024-2025 Irene Wu, University of Pennsylvania
- 2024 Reshana Ahmad, University of Pennsylvania
- 2023-2024 Trinity Kellarakos, University of Pennsylvania
- 2023-2024 Amelia Demopoulos, University of Pennsylvania

2022 Tynia Baker, University of Pennsylvania
2017-2018 Theodora Mautz, Harvard University
2015-2017 Maddie Snyder, Harvard University
2010 Oly Khowash, Princeton University
2009 Jan Kalkus, Princeton University

INVITED TALKS

Mind/Brain Institute seminar series, Johns Hopkins University, October 2025
Centre for Vision Research (CVR) seminar series, York University, September 2025
NPRC Consortium, Marmoset Neuroimaging subgroup, May 2025
University of California, Berkeley, Neuroscience Seminar Series, May 2025
CalTech, Division of Biology and Biological Engineering, August 2022
JHBI (Japanese meeting for Human Brain Imaging) Talk Series, January 2022
Giessen University, Seminar series: Current topics in Perception and Cognition, November 2021
Carnegie Mellon University, AI Seminar Series, December 2020
Nathan Klein Institute, Works in Progress Seminar Series, December 2020
University of California, Berkeley, Cognitive Neuroscience Colloquium, November 2020
Bar Illan University, Cognitive Neuroscience Lab, January 2019
University of Virginia, Department of Psychology Colloquium, March 2018
University of Pittsburgh, Department of Ophthalmology, March 2018
University of Minnesota, CMRR Seminar, January 2017
Massachusetts Institute of Technology, CBMM Special Seminar Series, December 2016

CONFERENCE PRESENTATIONS

<https://www.arcarolab.org/conferences.html>